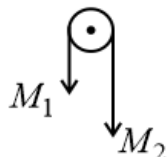


Q1. Two projectiles are thrown with same initial velocity making an angle of 45° and 30° with the horizontal respectively. The ratio of their respective ranges will be

- (1) $1:\sqrt{2}$ (2) $\sqrt{2}:1$
 (3) $2:\sqrt{3}$ (4) $\sqrt{3}:2$

Q2. Two masses M_1 and M_2 are tied together at the two ends of a light inextensible string that passes over a frictionless pulley. When the mass M_2 is twice that of M_1 , the acceleration of the system is a_1 . When the mass M_2 is thrice that of M_1 , the acceleration of The system is a_2 . The ratio $\frac{a_1}{a_2}$ will be



- (1) $\frac{1}{3}$ (2) $\frac{2}{3}$
 (3) $\frac{3}{2}$ (4) $\frac{1}{2}$

Q3. A ball of mass 0.15 kg hits the wall with its initial speed of 12 m s^{-1} and bounces back without changing its initial speed. If the force applied by the wall on the ball during the contact is 100 N , calculate the time duration of the contact of ball with the wall.

- (1) 0.018 s (2) 0.036 s
 (3) 0.009 s (4) 0.072 s

Q4. A body of mass 8 kg and another of mass 2 kg are moving with equal kinetic energy. The ratio of their respective momenta will be

- (1) $1:1$ (2) $2:1$
 (3) $1:4$ (4) $4:1$

Q5. A body is projected vertically upwards from the surface of earth with a velocity equal to one third of escape velocity. The maximum height attained by the body will be

(Take radius of earth = 6400 km and $g = 10 \text{ ms}^{-2}$)

- (1) 800 km (2) 1600 km
 (3) 2133 km (4) 4800 km

Q6. The area of cross section of the rope used to lift a load by a crane is $2.5 \times 10^{-4} \text{ m}^2$. The maximum lifting capacity of the crane is 10 metric tons. To increase the lifting capacity of the crane to 25 metric tons, the required area of cross section of the rope should be

(take $g = 10 \text{ ms}^{-2}$)

- (1) $6.25 \times 10^{-4} \text{ m}^2$ (2) $10 \times 10^{-4} \text{ m}^2$
 (3) $1 \times 10^{-4} \text{ m}^2$ (4) $1.67 \times 10^{-4} \text{ m}^2$

Q7. An ice cube of dimensions $60 \text{ cm} \times 50 \text{ cm} \times 20 \text{ cm}$ is placed in an insulation box of wall thickness 1 cm . The box keeping the ice cube at 0°C of temperature is brought to a room of temperature 40°C . The rate of melting of ice is approximately: (Latent heat of fusion of ice is $3.4 \times 10^5 \text{ J kg}^{-1}$ and thermal conducting of insulation wall is $0.05 \text{ W m}^{-1} ^\circ\text{C}^{-1}$)

(1) $61 \times 10^{-1} \text{ kg s}^{-1}$

(2) $61 \times 10^{-5} \text{ kg s}^{-1}$

(3) 208 kg s^{-1}

(4) $30 \times 10^{-5} \text{ kg s}^{-1}$

Q8. A gas has n degrees of freedom. The ratio of specific heat of gas at constant volume to the specific heat of gas at constant pressure will be

(1) $\frac{n}{n+2}$

(2) $\frac{n+2}{n}$

(3) $\frac{n}{2n+2}$

(4) $\frac{n}{n-2}$

Q9. A transverse wave is represented by $y = 2\sin\omega t - kx$ cm. The value of wavelength (in cm) for which the wave velocity becomes equal to the maximum particle velocity, will be

(1) 4π

(2) 2π

(3) π

(4) 2

Q10. Two uniformly charged spherical conductors A and B of radii 5 mm and 10 mm are separated by a distance of 2 cm . If the spheres are connected by a conducting wire, then in equilibrium condition, the ratio of the magnitudes of the electric fields at the surface of the sphere A and B will be

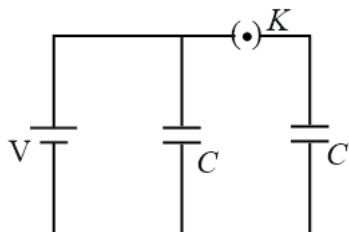
(1) $1:2$

(2) $2:1$

(3) $1:1$

(4) $1:4$

Q11. A source of potential difference V is connected to the combination of two identical capacitors as shown in the figure. When key K is closed, the total energy stored across the combination is E_1 . Now key K is opened and dielectric of dielectric constant 5 is introduced between the plates of the capacitors. The total energy stored across the combination is now E_2 . The ratio $\frac{E_1}{E_2}$ will be



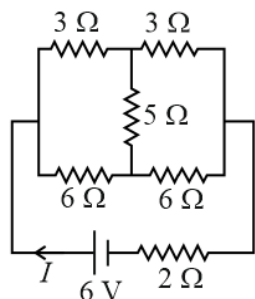
(1) $\frac{1}{10}$

(2) $\frac{2}{5}$

(3) $\frac{5}{13}$

(4) $\frac{5}{26}$

Q12. A battery of 6 V is connected to the circuit as shown below. The current I drawn from the battery is



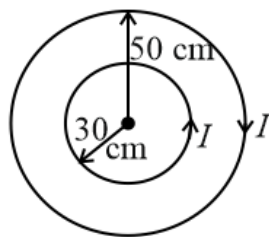
(1) 1 A

(2) 2 A

(3) $\frac{6}{11} \text{ A}$

(4) $\frac{4}{3} \text{ A}$

- Q13.** Two concentric circular loops of radii $r_1 = 30$ cm and $r_2 = 50$ cm are placed in $X - Y$ plane as shown in the figure. A current $I = 7$ A is flowing through them in the direction as shown in figure. The net magnetic moment of this system of two circular loops is approximately



- (1) $\frac{7}{2}\hat{k}$ A m² (2) $-\frac{7}{2}\hat{k}$ A m²
 (3) $7\hat{k}$ A m² (4) $-7\hat{k}$ A m²
- Q14.** A velocity selector consists of electric field $\vec{E} = E \hat{k}$ and magnetic field $\vec{B} = B \hat{j}$ with $B = 12$ mT. The value E required for an electron of energy 728 eV moving along the positive x-axis to pass undeflected is (Given, mass of electron = 9.1×10^{-31} kg)
- (1) 192 kV m⁻¹ (2) 192 mV m⁻¹
 (3) 9600 kV m⁻¹ (4) 16 kV m⁻¹
- Q15.** The oscillating magnetic field in a plane electromagnetic wave is given by $B_y = 5 \times 10^{-6} \sin 1000\pi x - 4 \times 10^8 t$ T. The amplitude of electric field will be
- (1) $15 \times 10^2 \text{ Vm}^{-1}$ (2) $5 \times 10^{-6} \text{ Vm}^{-1}$
 (3) $16 \times 10^{12} \text{ Vm}^{-1}$ (4) $4 \times 10^2 \text{ Vm}^{-1}$
- Q16.** Light travels in two media M_1 and M_2 with speeds 1.5×10^8 m s⁻¹ and 2.0×10^8 m s⁻¹ respectively. The critical angle between them is
- (1) $\tan^{-1} \frac{3}{4}$ (2) $\tan^{-1} \frac{2}{3}$
 (3) $\cos^{-1} \frac{3}{4}$ (4) $\sin^{-1} \frac{2}{3}$
- Q17.** A nucleus of mass M at rest splits into two parts having masses $\frac{M'}{3}$ and $\frac{2M'}{3}$ $M' < M$. The ratio of de Broglie wavelength of two parts will be
- (1) 1:2 (2) 2:1
 (3) 1:1 (4) 2:3
- Q18.** Mass numbers of two nuclei are in the ratio of 4:3. Their nuclear densities will be in the ratio of
- (1) 4:3 (2) $\frac{1}{3}$
 (3) 1:1 (4) $\frac{4}{3}$
- Q19.** The maximum and minimum voltage of an amplitude modulated signal are 60 V and 20 V respectively. The percentage modulation index will be
- (1) 0.5% (2) 50%
 (3) 2% (4) 30%

Q20. In a Vernier Caliper 10 divisions of Vernier scale is equal to the 9 divisions of main scale. When both jaws of Vernier calipers touch each other, the zero of the Vernier scale is shifted to the left of zero of the main scale and 4th Vernier scale division exactly coincides with the main scale reading. One main scale division is equal to 1 mm. While measuring diameter of a spherical body, the body is held between two jaws. It is now observed that zero of the Vernier scale lies between 30 and 31 divisions of main scale reading and 6th Vernier scale division exactly coincides with the main scale reading. The diameter of the spherical body will be:

- (1) 3.02 cm (2) 3.06 cm
(3) 3.10 cm (4) 3.20 cm

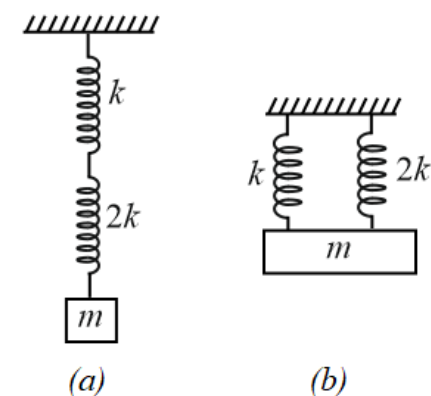
Q21. If $\vec{A} = 2\hat{i} + 3\hat{j} - \hat{k}$ m and $\vec{B} = \hat{i} + 2\hat{j} + 2\hat{k}$ m. The magnitude of component of vector $\vec{A}$ along vector $\vec{B}$ will be _____ m.

Q22. The radius of gyration of a cylindrical rod about an axis of rotation perpendicular to its length and passing through the center will be _____ m. Given, the length of the rod is $10\sqrt{3}$ m.

Q23. A uniform heavy rod of mass 20 kg. Cross sectional area 0.4 m^2 and length 20 m is hanging from a fixed support. Neglecting the lateral contraction, the elongation in the rod due to its own weight is $x \times 10^{-9}$ m. The value of x is _____.

(Given. Young's modulus $Y = 2 \times 10^{11} \text{ Nm}^{-2}$ and $g = 10 \text{ m s}^{-2}$)

Q24. As per given figures, two springs of spring constants K and $2K$ are connected to mass m . If the period of oscillation in figure (a) is 3 s, then the period of oscillation in figure (b) will be $\sqrt{x}$ s. The value of x is _____.

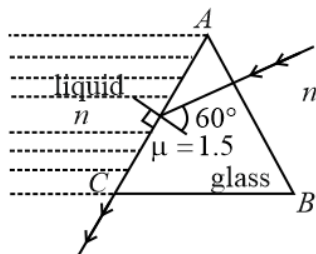


Q25. Three point charges of magnitude $5 \mu\text{C}$, $0.16 \mu\text{C}$ and $0.3 \mu\text{C}$ are located at the vertices A, B, C of a right angled triangle whose sides are $AB = 3 \text{ cm}$, $BC = 3\sqrt{2} \text{ cm}$ and $CA = 3 \text{ cm}$ and point A is the right angle corner. Charge at point A experiences _____ N of electrostatic force due to the other two charges.

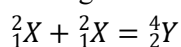
Q26. A potentiometer wire of length 300 cm is connected in series with a resistance 780Ω and a standard cell of emf 4 V. A constant current flows through potentiometer wire. The length of the null point for cell of emf 20mV is found to be 60 cm. The resistance of the potentiometer wire is _____ Ω .

Q27. In a coil of resistance 8Ω , the magnetic flux due to an external magnetic field varies with time as $\phi = \frac{2}{3}9 - t^2$. The value of total heat produced in the coil, till the flux becomes zero, will be _____ J.

- Q28.** In the given figure, the face AC of the equilateral prism is immersed in a liquid of refractive index n . For incident angle 60° at the side AC , the refracted light beam just grazes along face AC . The refractive index of the liquid $n = \frac{\sqrt{x}}{4}$. The value of x is _____.
(Given refractive index of glass = 1.5)

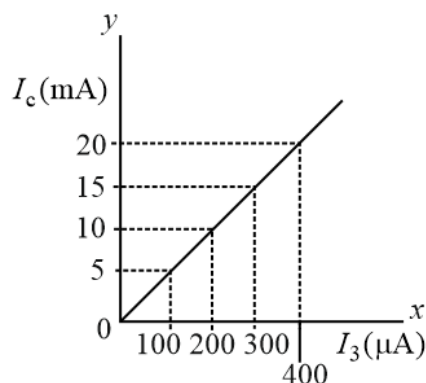


- Q29.** Two lighter nuclei combine to form a comparatively heavier nucleus by the relation given below:



The binding energies per nucleon 2_1X and 4_2Y are 1.1 MeV and 7.6 MeV respectively. The energy released in this process is _____ MeV.

- Q30.** The typical transfer characteristic of a transistor in CE configuration is shown in figure. A load resistor of 2 k Ω is connected in the collector branch of the circuit used. The input resistance of the transistor is 0.50 k Ω . The voltage gain of the transistor is



- Q31.** Hemoglobin contains 0.34% of iron by mass. The number of Fe atoms in 3.3 g of hemoglobin is (Given :

Atomic mass of Fe is 56u, N_A in $6.022 \times 10^{23} \text{ mol}^{-1}$)

- (1) 1.21×10^5 (2) 12.0×10^{16}
(3) 1.21×10^{20} (4) 3.4×10^{22}

- Q32.** The metal that has very low melting point and its periodic position is closer to a metalloid is

- (1) Al (2) Ga
(3) Se (4) In

- Q33.** Arrange the following in increasing order of their covalent character.

- (A) CaF_2
(B) CaCl_2
(C) CaBr_2
(D) CaI_2

Choose the correct answer from the options given below.

(1) $B < A < C < D$

(2) $A < B < C < D$

(3) $A < B < D < C$

(4) $A < C < B < D$

Q34. Class XII students were asked to prepare one litre of buffer solution of pH 8.26 by their chemistry teacher.

The amount of ammonium chloride to be dissolved by the student in 0.2M ammonia solution to make one litre of the buffer is (Given $pK_b \text{NH}_3 = 4.74$; Molar mass of $\text{NH}_3 = 17 \text{ g mol}^{-1}$. Molar mass of $\text{NH}_4\text{Cl} = 53.5 \text{ g mol}^{-1}$)

(1) 53.5 g

(2) 72.3 g

(3) 107 g

(4) 126g

Q35. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Phenolphthalein is a pH dependent indicator, remains colourless in acidic solution and gives pink colour in basic medium

Reason R : Phenolphthalein is a weak acid. It doesn't dissociate in basic medium.

In the light of the above statements, choose the most appropriate answer from the options given below

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is NOT the correct explanation of A.

(3) A is true but R is false

(4) A is false but R is true

Q36. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R. Assertion A : LiF is sparingly soluble in water. Reason R : The ionic radius of Li^+ ion is smallest among its group members, hence has least hydration enthalpy.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is NOT the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

Q37. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Boric acid is a weak acid

Reason R : Boric acid is not able to release H^+ ion on its own. It receives OH^- ion from water and releases H^+ ion.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both A and R are correct and R is the correct explanation of A

(2) Both A and R are correct but R is NOT the correct explanation of A

(3) A is correct but R is not correct

(4) A is not correct but R is correct

Q38. The correct decreasing order of priority of functional groups in naming an organic compound as per IUPAC system of nomenclature is

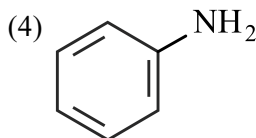
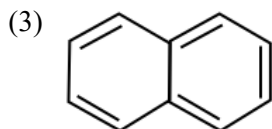
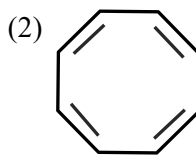
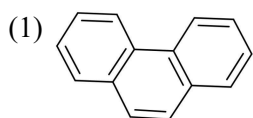
(1) $-\text{COOH} > -\text{CONH}_2 > -\text{COCl} > -\text{CHO}$

(2) $-\text{SO}_3\text{H} > -\text{COCl} > -\text{CONH}_2 > -\text{CN}$

(3) $-\text{COOR} > -\text{COCl} > \text{NH}_2 > \text{C}=\text{O}$

(4) $\text{COOH} > -\text{COOR} > -\text{CONH}_2 > -\text{COCl}$

Q39. Which of the following is not an example of benzenoid compound?



Q40. Match List I with List II

List-I**Pollutant**

- A Microorganisms
B Plant nutrients
C Toxic heavy metals
D Sediment

List-II**Source**

- I Strip mining
II Domestic sewage
III Chemical fertilizer
IV Chemical factory

Choose the correct answer from the options given below

(1) A - II, B - III, C - IV, D - I

(2) A - II, B - I, C - IV, D - III

(3) A - I, B - IV, C - II, D - III

(4) A - I, B - IV, C - III, D - II

Q41. At 30 °C, the half life for the decomposition of AB_2 is 200 s and is independent of the initial concentration of AB_2 . The time required for 80% of the AB_2 to decompose is (Given: $\log 2 = 0.30$; $\log 3 = 0.48$)

(1) 200s

(2) 323s

(3) 467s

(4) 532s

Q42. Given below are two statements: one is labelled as

Assertion A and the other is labelled as Reason R.

Assertion A : Finest gold is red in colour, as the size of the particles increases, it appears purple then blue and finally gold.

Assertion R : The colour of the colloidal solution depends on the wavelength of light scattered by the dispersed particles.

In the light of the above statements, choose the most appropriate answer from the options given below

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is NOT the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

Q43. The metal that is not extracted from its sulphide ore is

(1) Aluminium

(2) Iron

(3) Lead

(4) Zinc

Q44. The products obtained from a reaction of hydrogen peroxide and acidified potassium permanganate are

(1) Mn^{4+} , H_2O only

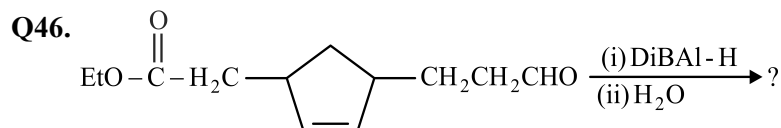
(2) Mn^{2+} , H_2O only

(3) Mn^{4+} , H_2O , O_2 only

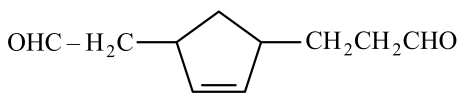
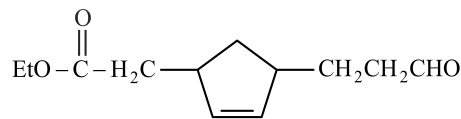
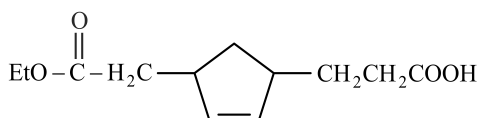
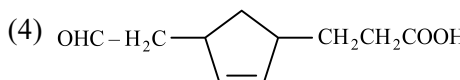
(4) Mn^{2+} , H_2O , O_2 only

Q45. The metal complex that is diamagnetic is (Atomic number : Fe, 26; Cu, 29)

- (1) K_3CuCN_4 (2) K_2CuCN_4
 (3) K_3FeCN_4 (4) K_4FeCl_6



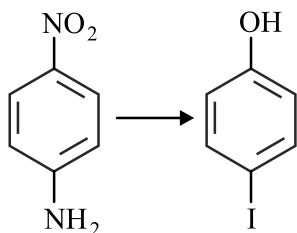
Consider the above reaction and predict the major product.

- (1)  (2) 
 (3)  (4) 

Q47. Hydrolysis of which compound will give carbolic acid?

- (1) Cumene (2) Benzenediazonium chloride
 (3) Benzal chloride (4) Ethylene glycol ketal

Q48. The correct sequential order of the reagents for the given reaction is



- (1) HNO_2 , Fe / H^+ , HNO_2 , KI, H_2O / H^+ (2) HNO_2 , KI, Fe / H^+ , HNO_2 , H_2O / warm
 (3) HNO_2 , KI, HNO_2 , Fe / H^+ , H_2O / H^+ (4) HNO_2 , Fe / H^+ , KI, HNO_2 , H_2O / warm

Q49. Vulcanization of rubber is carried out by heating a mixture of

- (1) isoprene and styrene (2) neoprene and sulphur
 (3) isoprene and sulphur (4) neoprene and styrene

Q50. Animal starch is the other name of

- (1) amylose (2) maltose
 (3) glycogen (4) amylopectin

Q51. Consider an imaginary ion ${}^{48}_{22}\text{X}^{3-}$. The nucleus contains 'a' % more neutrons than the number of electrons in the ion. The value of 'a' is

Q52. A 10 g mixture of hydrogen and helium is contained in a vessel of capacity 0.0125 m^3 at 6 bar and 27°C . The mass of helium in the mixture is g.

(Given : $R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$ (Atomic masses of H and He are 1u and 4u, respectively))

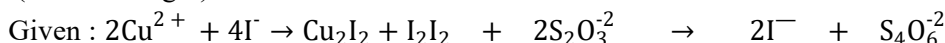
Q53. For the reaction $\text{H}_2\text{F}_2\text{g} \rightarrow \text{H}_2\text{g} + \text{F}_2\text{g}$

$$\Delta U = -59.6 \text{ kJ mol}^{-1} \text{ at } 27^\circ \text{C}$$

The enthalpy change for the above reaction is - ____ kJ mol^{-1} (nearest integer)

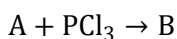
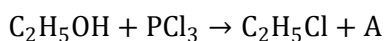
(Given : $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$).

Q54. 20 mL of 0.02 M hypo solution is used for the titration of 10 mL of copper sulphate solution, in the presence of excess of KI using starch as an indicator. The molarity of Cu^{2+} is found to be ____ $\times 10^{-2} \text{ M}$ (nearest integer)



Q55. The elevation in boiling point for 1 molal solution of non-volatile solute A is 3 K. The depression in freezing point for 2 molal solution of A in the same solvent is 6 K. The ratio of K_b and K_f i.e., K_b / K_f is 1:X. The value of X is

Q56. The number of non-ionisable protons present in the product B obtained from the following reaction is ____



Q57. The spin-only magnetic moment value of the compound with strongest oxidizing ability among MnF_4 , MnF_3 and MnF_2 is ____ B. M (nearest integer)

Q58. Total number of isomers (including stereoisomers) obtain on monochlorination of methylcyclohexane is

Q59. A 100 mL solution of $\text{CH}_3\text{CH}_2\text{MgBr}$ on treatment with methanol produces 2.24 mL of a gas at STP. The weight of gas produced is ____ mg (nearest integer)

Q60. How many of the following drugs is/are example(s) of broad spectrum antibiotic? Ofloxacin, Penicillin G, Terpineol, Salvarsan

Q61. The minimum value of the sum of the squares of the roots of $x^2 + 3 - ax = 2a - 1$ is

(1) 6

(2) 4

(3) 5

(4) 8

Q62. If $z = x + iy$ satisfies $z - 2 = 0$ and $z - i - z + 5i = 0$, then

(1) $x + 2y - 4 = 0$

(2) $x^2 + y - 4 = 0$

(3) $x + 2y + 4 = 0$

(4) $x^2 - y + 3 = 0$

Q63. $\sum_{i,j=0}^n {}^nC_i {}^nC_j$ is equal to

(1) $2^{2n} - {}^{2n}C_n$

(2) $2^{2n-1} - {}^{2n-1}C_{n-1}$

(3) $2^{2n} - \frac{1}{2} {}^{2n}C_n$

(4) $2^{n-1} + {}^{2n-1}C_n$

Q64. Let the abscissae of the two points P and Q on a circle be the roots of $x^2 - 4x - 6 = 0$ and the ordinates of P and Q be the roots of $y^2 + 2y - 7 = 0$. If PQ is a diameter of the circle $x^2 + y^2 + 2ax + 2by + c = 0$, then the value of $a + b - c$ is

(1) 12

(2) 13

(3) 14

(4) 16

Q65. The equation of a common tangent to the parabolas $y = x^2$ and $y = -x - 2^2$ is

- (1) $y = 4x - 2$ (2) $y = 4x - 1$
 (3) $y = 4x + 1$ (4) $y = 4x + 2$

Q66. The acute angle between the pair of tangents drawn to the ellipse $2x^2 + 3y^2 = 5$ from the point 1, 3 is

- (1) $\tan^{-1} \frac{16}{7\sqrt{5}}$ (2) $\tan^{-1} \frac{24}{7\sqrt{5}}$
 (3) $\tan^{-1} \frac{32}{7\sqrt{5}}$ (4) $\tan^{-1} \frac{3+8\sqrt{5}}{35}$

Q67. If the line $x - 1 = 0$, is a directrix of the hyperbola $kx^2 - y^2 = 6$, then the hyperbola passes through the point

- (1) $-2\sqrt{5}, 6$ (2) $-\sqrt{5}, 3$
 (3) $\sqrt{5}, -2$ (4) $2\sqrt{5}, 3\sqrt{6}$

Q68. Let $\beta = \lim_{x \rightarrow 0} \frac{\alpha x - e^{3x} - 1}{\alpha x e^{3x} - 1}$ for some $\alpha \in \mathbb{R}$. Then the value of $\alpha + \beta$ is:

- (1) $\frac{14}{5}$ (2) $\frac{3}{2}$
 (3) $\frac{5}{2}$ (4) $\frac{7}{2}$

Q69. Negation of the Boolean expression $p \leftrightarrow q \rightarrow p$ is

- (1) $\sim p \wedge q$ (2) $p \wedge \sim q$
 (3) $\sim p \vee \sim q$ (4) $\sim p \wedge \sim q$

Q70. Let $A = \begin{pmatrix} 1 & 9^2 & -10^2 & 11^2 \\ 1 & 12^2 & 13^2 & -14^2 \\ 1 & -15^2 & 16^2 & 17^2 \end{pmatrix}$ and $B = \begin{pmatrix} 9^2 & -10^2 & 11^2 \\ 12^2 & 13^2 & -14^2 \\ -15^2 & 16^2 & 17^2 \end{pmatrix}$, then the value of $A'BA$ is;

- (1) 1224 (2) 1042
 (3) 540 (4) 539

Q71. If $0 < x < \frac{1}{\sqrt{2}}$ and $\frac{\sin^{-1}x}{\alpha} = \frac{\cos^{-1}x}{\beta}$, then a value of $\sin \frac{2\pi\alpha}{\alpha+\beta}$ is

- (1) $4\sqrt{1-x^2} - 2x^2$ (2) $4x\sqrt{1-x^2} - 2x^2$
 (3) $2x\sqrt{1-x^2} - 4x^2$ (4) $4\sqrt{1-x^2} - 4x^2$

Q72. The value of $\log_e 2 \frac{d}{dx} \log_{\cos x} \operatorname{cosec} x$ at $x = \frac{\pi}{4}$ is

- (1) $-2\sqrt{2}$ (2) $2\sqrt{2}$
 (3) -4 (4) 4

Q73. Let P and Q be any points on the curves $x^2 + y^2 = 1$ and $y = x^2$, respectively. The distance between P and Q is minimum for some value of the abscissa of P in the interval

- (1) $0, \frac{1}{4}$ (2) $\frac{1}{2}, \frac{3}{4}$
 (3) $\frac{1}{4}, \frac{1}{2}$ (4) $\frac{3}{4}, 1$

Q74. If the maximum value of a , for which the function $f_a x = \tan^{-1} 2x - 3ax + 7$ is non-decreasing in $-\frac{\pi}{6}, \frac{\pi}{6}$, is $\bar{a}$, then $f_{\bar{a}} \frac{\pi}{8}$ is equal to

- (1) $8 - \frac{9\pi}{49 + \pi^2}$ (2) $8 - \frac{4\pi}{94 + \pi^2}$
 (3) $8 - \frac{1 + \pi^2}{9 + \pi^2}$ (4) $8 - \frac{\pi}{4}$

Q75. The integral $\int \frac{1 - \frac{1}{\sqrt{3}}\cos x - \sin x}{1 + \frac{2}{\sqrt{3}}\sin 2x} dx$ is equal to

(1) $\frac{1}{2} \log_e \frac{\tan \frac{x}{2} + \frac{\pi}{12}}{\frac{x}{2} + \frac{\pi}{6}} + C$

(2) $\log_e \frac{\tan \frac{x}{2} + \frac{\pi}{6}}{\frac{x}{2} + \frac{\pi}{3}} + C$

(3) $\frac{1}{2} \log_e \frac{\tan \frac{x}{2} + \frac{\pi}{6}}{\frac{x}{2} + \frac{\pi}{3}} + C$

(4) $\frac{1}{2} \log_e \frac{\tan \frac{x}{2} - \frac{\pi}{12}}{\tan \frac{x}{2} - \frac{\pi}{6}} + C$

Q76. $\int_0^{20\pi} \sin x + \cos x^2 dx$ is equal to:

(1) $10\pi + 4$

(2) $10\pi + 2$

(3) $20\pi - 2$

(4) $20\pi + 2$

Q77. The area bounded by the curves $y = x^2 - 1$ and $y = 1$ is

(1) $\frac{2}{3}\sqrt{2} + 1$

(2) $\frac{4}{3}\sqrt{2} - 1$

(3) $2\sqrt{2} - 1$

(4) $\frac{8}{3}\sqrt{2} - 1$

Q78. Let the solution curve $y = fx$ of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2 - 1} = \frac{x^4 + 2x}{\sqrt{1 - x^2}}$, $x \in -1, 1$ pass through the

origin. Then $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} fxdx$ is equal to

(1) $\frac{\pi}{3} - \frac{1}{4}$

(2) $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$

(3) $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$

(4) $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

Q79. A vector $\vec{a}$ is parallel to the line of intersection of the plane determined by the vectors $\hat{i}, \hat{i} + \hat{j}$ and the plane determined by the vectors $\hat{i} - \hat{j}, \hat{i} + \hat{k}$. The obtuse angle between $\vec{a}$ and the vector $\vec{b} = \hat{i} - 2\hat{j} + 2\hat{k}$ is

(1) $\frac{3\pi}{4}$

(2) $\frac{2\pi}{3}$

(3) $\frac{4\pi}{5}$

(4) $\frac{5\pi}{6}$

Q80. Let X be a binomially distributed random variable with mean 4 and variance $\frac{4}{3}$. Then $54 \cdot P(X \leq 2)$ is equal to

(1) $\frac{73}{146}$

(2) $\frac{146}{27}$

(3) $\frac{146}{81}$

(4) $\frac{126}{81}$

Q81. Numbers are to be formed between 1000 and 3000, which are divisible by 4, using the digits 1, 2, 3, 4, 5 and 6 without repetition of digits. Then the total number of such numbers is _____.

Q82. If $\sum_{k=1}^{10} \frac{k}{k^4 + k^2 + 1} = \frac{m}{n}$, where m and n are co-prime, then $m + n$ is equal to

Q83. Different A.P.'s are constructed with the first term 100, the last term 199, And integral common differences.

The sum of the common differences of all such, A.P.'s having at least 3 terms and at most 33 terms is.

Q84. If the sum of solutions of the system of equations $2\sin^2\theta - \cos 2\theta = 0$ and $2\cos^2\theta + 3\sin\theta = 0$ in the interval $0, 2\pi$ is $k\pi$, then k is equal to _____.

Q85. The mean and standard deviation of 40 observations are 30 and 5 respectively. It was noticed that two of these observations 12 and 10 were wrongly recorded. If σ is the standard deviation of the data after omitting the two wrong observations from the data, then $38\sigma^2$ is equal to _____.

Q86. Let $A = 1, 2, 3, 4, 5, 6, 7$ and $B = 3, 6, 7, 9$. Then the number of elements in the set $C \subseteq A: C \cap B \neq \phi$ is _____.

Q87. The number of matrices $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where $a, b, c, d \in -1, 0, 1, 2, 3, \dots, 10$, such that $A = A^{-1}$, is _____.

Q88. Suppose $y = yx$ be the solution curve to the differential equation $\frac{dy}{dx} - y = 2 - e^{-x}$ such that $\lim_{x \rightarrow \infty} yx$ is finite.

If a and b are respectively the x - and y - intercept of the tangent to the curve at $x = 0$, then the value of $a - 4b$ is equal to _____.

Q89. The largest value of a , for which the perpendicular distance of the plane containing the lines $\vec{r} = \hat{i} + \hat{j} + \lambda \hat{i} + a \hat{j} - \hat{k}$ and $\vec{r} = \hat{i} + \hat{j} + \mu \hat{i} + \hat{j} - a \hat{k}$ from the point $2, 1, 4$ is $\sqrt{3}$, is _____.

Q90. The plane passing through the line $L: l \cdot x - y + 31 - l \cdot z = 1, x + 2y - z = 2$ and perpendicular to the plane $3x + 2y + z = 6$ is $3x - 8y + 7z = 4$. If θ is the acute angle between the line L and the y -axis, then $415 \cos^2 \theta$ is equal to _____.

ANSWER KEYS

1. (3)	2. (2)	3. (2)	4. (2)	5. (1)	6. (1)	7. (2)	8. (1)
9. (1)	10. (2)	11. (3)	12. (1)	13. (2)	14. (1)	15. (4)	16. (1)
17. (3)	18. (3)	19. (2)	20. (3)	21. (2)	22. (5)	23. (25)	24. (2)
25. (17)	26. (20)	27. (2)	28. (27)	29. (26)	30. (200)	31. (3)	32. (2)
33. (2)	34. (3)	35. (3)	36. (3)	37. (1)	38. (2)	39. (2)	40. (1)
41. (3)	42. (1)	43. (1)	44. (4)	45. (1)	46. (1)	47. (2)	48. (2)
49. (3)	50. (3)	51. (4)	52. (8)	53. (57)	54. (4)	55. (1)	56. (2)
57. (5)	58. (12)	59. (3)	60. (1)	61. (1)	62. (3)	63. (1)	64. (1)
65. (2)	66. (2)	67. (3)	68. (3)	69. (4)	70. (4)	71. (2)	72. (4)
73. (3)	74. (1)	75. (1)	76. (4)	77. (4)	78. (2)	79. (1)	80. (2)
81. (30)	82. (166)	83. (53)	84. (3)	85. (238)	86. (112)	87. (50)	88. (3)
89. (2)	90. (125)						